

Counting LCM Arrays

Time limit: 1.00 s

Memory limit: 512 MB

Given two integers n and k , your task is to count the number of arrays $a_1, a_2, \dots, a_n$ of positive integers where $\text{lcm}(a_i, a_{i+1}) = k$ for all $1 \leq i < n$.

Input

The first line has an integer t : the number of test cases.

The next t lines have two integers n and k : the length of the array and the value of lcm.

Output

Print t integers: the answer to each test case modulo $10^9 + 7$.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n \leq 10^9$
- $1 \leq k \leq 10^9$

Example

Input	Output
3 3 4 4 6 1337 42	11 64 602746233

Explanation: The arrays for the first test case are $[1, 4, 1]$, $[1, 4, 2]$, $[1, 4, 4]$, $[2, 4, 1]$, $[2, 4, 2]$, $[2, 4, 4]$, $[4, 1, 4]$, $[4, 2, 4]$, $[4, 4, 1]$, $[4, 4, 2]$ and $[4, 4, 4]$.